I DEWA MADE DHARMA PUTRA SANTIKA

AIoT Software Engineer • Information Technology Student

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PROFILE

I'm an Information Technology student at ITB STIKOM Bali, focused on artificial intelligence, Internet of Things (IoT), and robotics, with a strong interest in AI-driven software development. I have experience building web and mobile products (React Native, Next.js), integrating IoT devices (MQTT, Node-RED, Raspberry Pi/ESP32), contributing to applied research, and leading a student organization. I work collaboratively, analytically, and adaptively in dynamic environments, with an approach to interface design that stays clean and human-centered.

EXPERIENCE

IoT Development Intern, PT Bima Sakti Sanjaya, Denpasar, Bali *Mar 2026 - Jul 2026*

- Contributed to AgriSense, a smart-farming IoT platform: built the real-time web dashboard in Next.js (live MQTT gauges, charts, threshold alerts, CSV export) and developed features of the React Native mobile app (sensor dashboard, history charts, activity log, remote actuator control).
- Implemented MQTT data subscriptions and validated sensor telemetry end to end using Node-RED, integrating field sensors with a Raspberry Pi edge device.
- Integrated Google Gemini with a rule-based validation step to generate crop recommendations from live sensor data.
- Designed a responsive UI and data visualizations for water quality (EC/TDS), air (temperature/humidity), and soil (moisture/pH/NPK) monitoring to support data-driven decisions.
- Joined senior engineers on site visits to inspect and maintain RTU Smart Panel installations, including troubleshooting, wiring checks, and connectivity verification.
- Structured sensor data for future AI use cases such as anomaly detection and predictive irrigation analytics.

ORGANIZATIONAL EXPERIENCE

General Chair, Robotics and Embedded Systems Research Group (RADE), ITB STIKOM Bali *Jan 2025 - Feb 2026*

- Led cross-functional teams to align project goals with organizational objectives, improving delivery efficiency and stakeholder satisfaction.
- Established rigorous quality assurance processes to ensure compliance with research standards and the reliability of deliverables.
- Coordinated research, training, seminars, and competitions in IoT, robotics, and AI, helping RADE earn recognition at the national level.

FEATURED PROJECTS

NADI: Navigasi Adaptif Distribusi Intelijen, Team Lead, GEMASTIK XIX 2026 Competition Proposal

- Led a 3 person team designing a smart tourism system that helps travelers in Bali make travel decisions based on actual and predicted conditions at arrival time (arrival time context intelligence), addressing uneven tourist distribution and overtourism at popular destinations.
- Designed a tiered adaptive itinerary mechanism (keep, reroute, reschedule, reorder, replace) that adjusts travel plans in response to changing traffic, destination crowding, incidents, and weather, while keeping the final decision with the user.
- Designed the intelligence layer architecture, including YOLO-based computer vision for CCTV/ATCS data, destination crowd forecasting, and a tourism flow distribution recommendation system, integrated with Google Maps Platform, OpenWeather, and the Gemini API.
- Built a working prototype of the NADI Control Center, a city operator dashboard for live monitoring, incident verification, tourism distribution, and policy evaluation.

SIGMA: Smart Intelligent Guidance for Mobility Automation, Team Lead, GEMASTIK XVIII 2025 Competition Proposal

- An IoT and AI based smart traffic signal system using YOLOv8 to detect vehicles and automatically adjust green light duration to reduce urban congestion, running on edge computing (Raspberry Pi) with an analytics dashboard.

AgriSense Mobile, React Native Mobile App

- Real-time farm and greenhouse monitoring app with live sensor readings, remote actuator control, weather insights, AI-based recommendations, and historical data visualization in a clean, modern interface.

COASTHER, Smart Boarding House IoT System

- An IoT system for smart boarding houses that monitors electricity consumption (ESP32, PZEM-004T) and water usage (YF-S201) in real time over MQTT/HiveMQ.
- Built the monitoring dashboard in Next.js and supported a companion React Native app.

Research I: Hybrid K-NN for Dangdut Music Sub-Genre Classification, Published, Journal of Applied Informatics and Computing (JAIC) *Aug 2025*

- Contributed as a researcher applying a Genetic Algorithm (GA) optimized Hybrid K-Nearest Neighbor algorithm to classify dangdut music sub-genres (classic, rock, koplo), using six spectral features (centroid, skewness, rolloff, kurtosis, spread, flatness) extracted from 483 audio segments of 30 seconds each.
- GA was used to optimize the k parameter with a population size of 10, 15 generations, a 4-bit chromosome length, and 3-fold validation during optimization, before the final model was evaluated using 5-fold cross-validation.
- The Hybrid K-NN model reached its highest accuracy of 74.31% at k=4 (average 72.04%, standard deviation 2.22%, with average precision, recall, and F1-score around 0.72), outperforming conventional K-NN (68.75%), Decision Tree (61.11%), and SVM with an ECOC scheme (54.86%).
- Published in the Journal of Applied Informatics and Computing (JAIC), Vol. 9 No. 4 (August 2025), SINTA 3 indexed. Read the article here (DOI: 10.30871/jaic.v9i4.9702).

Research II: MFCC Feature Performance Analysis for Dangdut Sub-Genre Classification, Published, Journal of Applied Informatics and Computing (JAIC), Follow-Up Study *Jun 2026*

- Extended the earlier study by applying Mel-Frequency Cepstral Coefficient features (13 MFCC coefficients) across 730 audio samples, comparing the performance of conventional K-NN, GA-optimized Hybrid K-NN, Support Vector Machine, and Decision Tree.
- Performed data normalization and PCA analysis to validate class separability, split the data into training and testing sets (70:30), and evaluated the models using accuracy, precision, recall, F1-score, confusion matrix, and cross-validation.
- The GA-optimized Hybrid K-NN model achieved the best accuracy of 98.90%, with consistently strong cross-validation results and low standard deviation, indicating good generalization without significant overfitting.
- Published in the Journal of Applied Informatics and Computing (JAIC), Vol. 10 No. 3 (June 2026), SINTA 3 indexed. Read the article here (DOI: 10.30871/jaic.v10i3.12760).

EDUCATION

Bachelor of Information Technology, ITB STIKOM Bali *Oct 2023 - May 2027 (expected)*

Focused on artificial intelligence, IoT, and robotics, with active involvement in campus research and student organizations.

Student Exchange Program, Valaya Alongkorn Rajabhat University (VRU), Thailand, 2025: Cloud Computing, Project Management, IT Research Methodology.

SKILLS

Programming Languages: JavaScript/TypeScript, Python
Frontend & Mobile: React Native, Next.js, Tailwind CSS
AI & Machine Learning: TensorFlow, OpenCV, YOLOv8, Feature Engineering
IoT & Embedded Systems: Arduino/ESP32, Raspberry Pi, RTU, Node-RED, MQTT
Data & Infrastructure: MySQL, PostgreSQL, phpMyAdmin, Firebase, GitHub
Soft Skills: Leadership & Team Management, Public Speaking, Strategic Planning, Problem Solving, Collaboration, Cross-Cultural Communication

CERTIFICATIONS

Project Integration Methodology of Excellence, Logical Operations *Aug 2026*

Credential License ID 193701929.

Android Development Associate, Logical Operations *Feb 2026*

Credential ID 181045116 • TypeScript, CSS, and related skills.

Python Programming Associate, Logical Operations *Feb 2025*

Python (Programming Language).

LANGUAGES

Indonesian: Native
English: B2 (Upper Intermediate)